

(1.1) Dokažte, že v libovolném komplexním vektorovém prostoru V se skalárním součinem $\langle \cdot, \cdot \rangle$ platí (pro libovolné $u, v \in V$)

$$\operatorname{Re}(\langle u, v \rangle) = \frac{1}{4}(\|u + v\|^2 - \|u - v\|^2),$$

$$\operatorname{Im}(\langle u, v \rangle) = \frac{1}{4i}(\|u - iv\|^2 - \|u + iv\|^2),$$

kde $\operatorname{Re}(x)$ značí reálnou a $\operatorname{Im}(x)$ imaginární část komplexního čísla x . (Pro inspiraci vizte tvrzení 8.32 ve skriptech.)

$$\operatorname{Re}(\langle u, v \rangle) = \frac{1}{4}(\|u + v\|^2 - \|u - v\|^2).$$

Nejprve upravíme:

$$\begin{aligned} \|u + v\|^2 &\stackrel{1}{=} \langle u + v, u + v \rangle \stackrel{SL2}{=} \langle u + v, u \rangle + \langle u + v, v \rangle = \\ &\stackrel{SSS}{=} \overline{\langle u, u + v \rangle} + \overline{\langle v, u + v \rangle} \stackrel{SL2}{=} \overline{\langle u, u \rangle} + \overline{\langle u, v \rangle} + \overline{\langle v, u \rangle} + \overline{\langle v, v \rangle} = \\ &\stackrel{SSS}{=} \langle u, u \rangle + \overline{\langle u, v \rangle} + \langle u, v \rangle + \langle v, v \rangle \end{aligned}$$

$$\begin{aligned} \|u - v\|^2 &\stackrel{1}{=} \langle u - v, u - v \rangle \stackrel{SL2}{=} \langle u - v, u \rangle + \langle u - v, -v \rangle \stackrel{SL1}{=} \\ &= \langle u - v, u \rangle - \langle u - v, v \rangle \stackrel{SSS}{=} \overline{\langle u, u - v \rangle} - \overline{\langle v, u - v \rangle} = \\ &\stackrel{SL2}{=} \overline{\langle u, u \rangle} + \overline{\langle u, -v \rangle} - \overline{\langle v, u \rangle} - \overline{\langle v, -v \rangle} = \\ &\stackrel{SL1}{=} \overline{\langle u, u \rangle} - \overline{\langle u, v \rangle} - \overline{\langle v, u \rangle} + \overline{\langle v, v \rangle} \stackrel{SSS}{=} \langle u, u \rangle - \overline{\langle u, v \rangle} - \langle u, v \rangle + \langle v, v \rangle \end{aligned}$$

Dosadíme je zpátky do výrazu.

$$\begin{aligned} \frac{1}{4}(\langle u, u \rangle + \overline{\langle u, v \rangle} + \langle u, v \rangle + \langle v, v \rangle - (\langle u, u \rangle - \overline{\langle u, v \rangle} - \langle u, v \rangle + \langle v, v \rangle)) = \\ = \frac{1}{4}(2\overline{\langle u, v \rangle} + 2\langle u, v \rangle) = \frac{1}{2}(\overline{\langle u, v \rangle} + \langle u, v \rangle) \end{aligned}$$

Protože $x + \bar{x} = 2\operatorname{Re}(x)$, tak $(\langle u, v \rangle + \overline{\langle u, v \rangle}) = 2\operatorname{Re}(\langle u, v \rangle)$ a $\frac{1}{2}(\langle u, v \rangle + \overline{\langle u, v \rangle}) = \operatorname{Re}(\langle u, v \rangle)$. Strany se rovnají!

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$$\operatorname{Im}(\langle u, v \rangle) = \frac{1}{4} (\|u - iv\|^2 - \|u + iv\|^2)$$

zачинне úpravou:

$$\begin{aligned} \|u - iv\|^2 &\stackrel{1}{=} \langle u - iv, u - iv \rangle \stackrel{SL2}{=} \langle u - iv, u \rangle + \langle u - iv, -iv \rangle \stackrel{SL1}{=} \langle u - iv, u \rangle - \langle u - iv, iv \rangle = \\ &\stackrel{SSS}{=} \overline{\langle u, u - iv \rangle} - \overline{\langle iv, u - iv \rangle} \stackrel{SL2}{=} \overline{\langle u, u \rangle} + \overline{\langle u, -iv \rangle} - \overline{\langle iv, u \rangle} - \overline{\langle iv, -iv \rangle} = \\ &\stackrel{SL1}{=} \overline{\langle u, u \rangle} + \overline{(-i) \langle u, v \rangle} - \overline{\langle iv, u \rangle} - \overline{(-i) \langle iv, v \rangle} = \\ &\stackrel{SSS}{=} \langle u, u \rangle + i \overline{\langle u, v \rangle} - \langle u, iv \rangle - i \langle v, iv \rangle = \\ &\stackrel{SL1}{=} \langle u, u \rangle + i \overline{\langle u, v \rangle} - i \langle u, v \rangle - i^2 \langle v, v \rangle = \langle u, u \rangle + i \overline{\langle u, v \rangle} - i \langle u, v \rangle + \langle v, v \rangle \end{aligned}$$

$$\begin{aligned} \|u + iv\|^2 &\stackrel{1}{=} \langle u + iv, u + iv \rangle \stackrel{SL2}{=} \langle u + iv, u \rangle + \langle u + iv, v \rangle \stackrel{SSS}{=} \overline{\langle u, u + iv \rangle} + \overline{\langle v, u + iv \rangle} = \\ &\stackrel{SL2}{=} \overline{\langle u, u \rangle} + \overline{\langle u, iv \rangle} + \overline{\langle iv, u \rangle} + \overline{\langle iv, v \rangle} \stackrel{SL1}{=} \overline{\langle u, u \rangle} + \overline{(i) \langle u, v \rangle} + \\ &+ \overline{\langle iv, u \rangle} + \overline{(i) \langle iv, v \rangle} \stackrel{SSS}{=} \langle u, u \rangle - i \overline{\langle u, v \rangle} + \langle u, iv \rangle - i \langle v, iv \rangle = \\ &\stackrel{SL1}{=} \langle u, u \rangle - i \overline{\langle u, v \rangle} + i \langle u, v \rangle - i^2 \langle v, v \rangle = \\ &= \langle u, u \rangle - i \overline{\langle u, v \rangle} + i \langle u, v \rangle + \langle v, v \rangle \end{aligned}$$

Dosaďme do výrazu

$$\begin{aligned} \frac{1}{4} (\|u - iv\|^2 - \|u + iv\|^2) &= \frac{1}{4} (\langle u, u \rangle + i \overline{\langle u, v \rangle} - i \langle u, v \rangle + \langle v, v \rangle - \\ &- (\langle u, u \rangle - i \overline{\langle u, v \rangle} + i \langle u, v \rangle + \langle v, v \rangle)) = \\ &= \frac{1}{4} (2i \overline{\langle u, v \rangle} - 2i \langle u, v \rangle) = \frac{1}{2} i (\overline{\langle u, v \rangle} - \langle u, v \rangle) = \end{aligned}$$

Pro $x = a + bi$ platí $\overline{x} - x = a - bi - a - bi = -2bi$ a

b představuje $\operatorname{Im}(x)$. Naš výraz je tedy roven

$$= \frac{1}{2} i (-2 \operatorname{Im}(\langle u, v \rangle) i) = -i^2 \operatorname{Im}(\langle u, v \rangle) = \operatorname{Im}(\langle u, v \rangle)$$

Strany se rovnají!

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